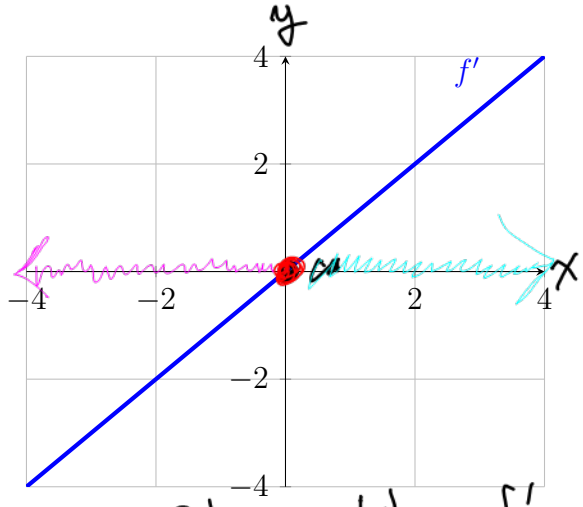
Goal: Recover information about a function f from the graph of f'.

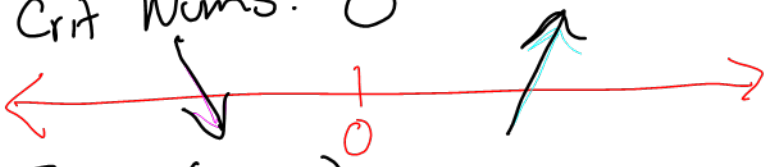
1 Implementing the I/D and 1st Derivative Test

Problem 1. *Use the graph of f' to determine the following about f:*



(a) Intervals of increase/decrease

Crit Nums: 0



Inc: $(0, \infty)$

Dec: $(-\infty, 0)$

(b) Positions of relative max/mins

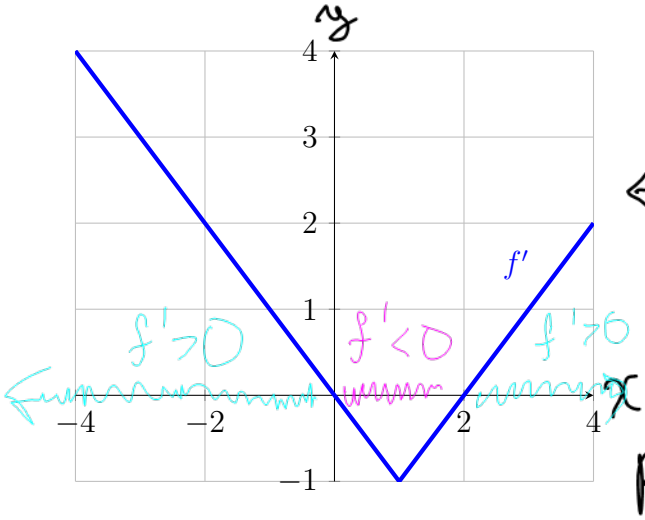
Rel. min at $x = 0$ by the 1st derivative test

Crit Nums ← $f' = 0$ ← where f' touches the x-axis

$f' > 0$ ← Graph is above the x-axis

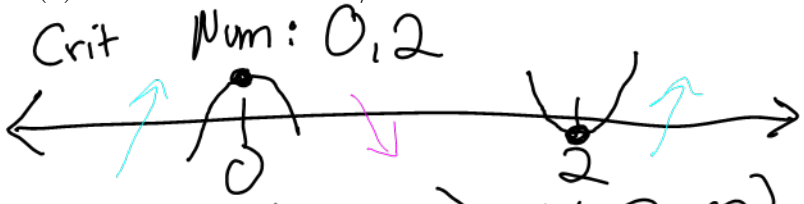
$f' < 0$ ← " " below " "

Problem 2. *Use the graph of f' to determine the following about f:*



(a) Intervals of increase/decrease

Crit Num: 0, 2



Inc: $(-\infty, 0) \cup (2, \infty)$

Dec: $(0, 2)$

(b) Positions of relative max/mins

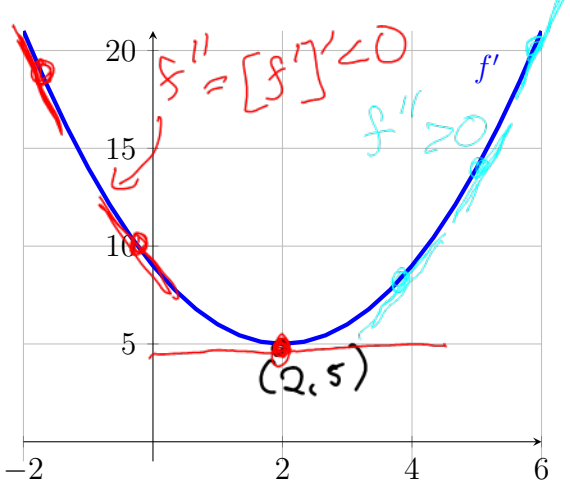
Rel. Max at $x = 0$

Rel. min at $x = 2$

by the 1st Derivative Test

2 Implementing the Concavity and 2nd Derivative Test

Problem 3. *Use the graph of f' to determine the following about f:* $[f'']=$



(a) When f is concave up VS. concave down



CD on $(-\infty, 2)$

CU on $(2, \infty)$

(b) Position of any inflection points

$f'' = [f']' \approx$ Slope of the Tangent Line of f'

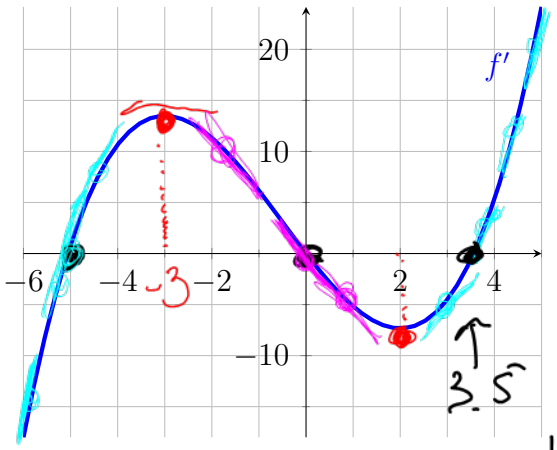
IP at $x = 2$

$f'' = 0 \leftarrow$ Slope Flatlines

$f'' < 0 \leftarrow$ Slopes are negative

$\leftarrow$ Lines "Point Downward"

Problem 4. *Use the graph of f' to determine the following about f:*



(a) When f is concave up VS. concave down



CU on $(-\infty, -3) \cup (2, \infty)$

CD on $(-3, 2)$

(b) Position of any inflection points

(c) Crit nums: $x = -5, 0, 3.5$

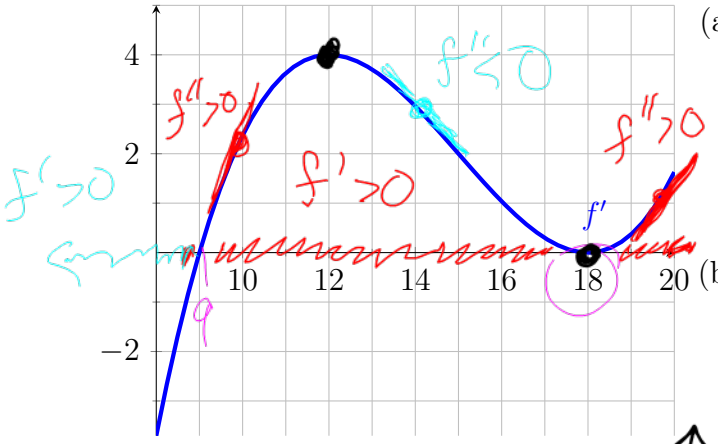
$f''(-5) > 0 \Rightarrow$ Rel. min at $x = -5$

$f''(0) < 0 \Rightarrow$ Rel. Max at $x = 0$

Both $x = -3$ and $x = 2$ are locations of IPs

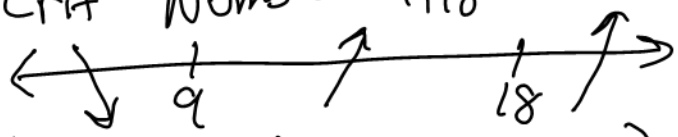
3 Putting Everything Together

Problem 5. *Use the graph of f' to determine the following about f:*



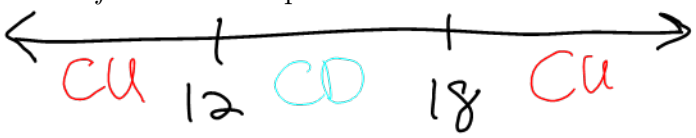
(a) Intervals of Increase/Decrease

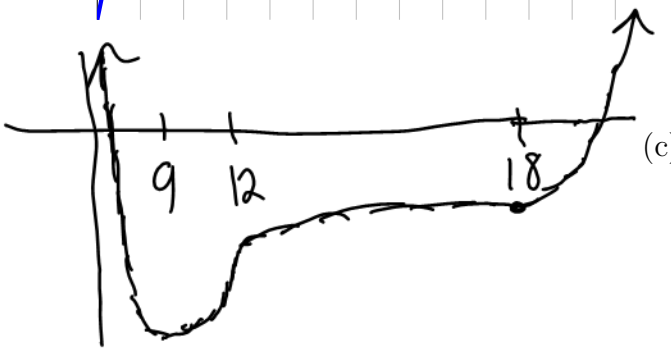
Crit Nums: 9, 18



Dec: $(-\infty, 9)$ Inc: $(9, \infty)$

(b) When f is concave up VS concave down



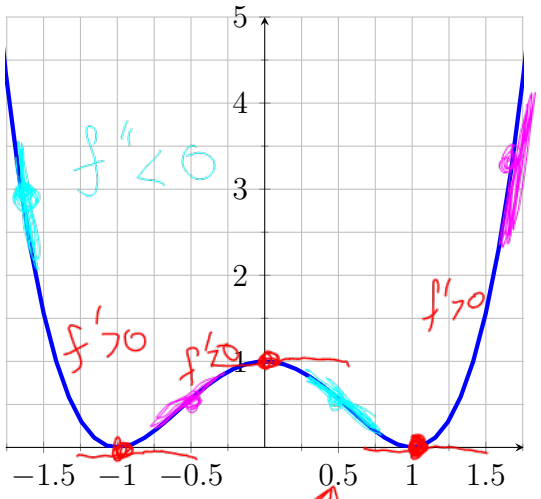


(c) State the locations of any relative max/mins as well as inflection points

Rel min: $x = 9$ Rel max: None

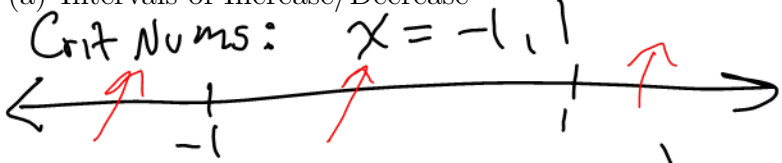
IPs: At $x = 12, 18$

Problem 6. *Use the graph of f' to determine the following about f:*



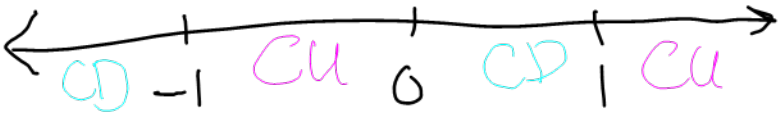
(a) Intervals of Increase/Decrease

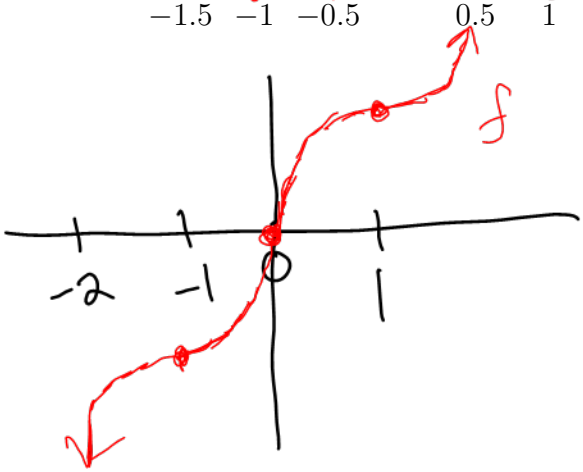
Crit Nums: $x = -1, 1$



Inc on $(-\infty, \infty)$

(b) When f is concave up VS concave down





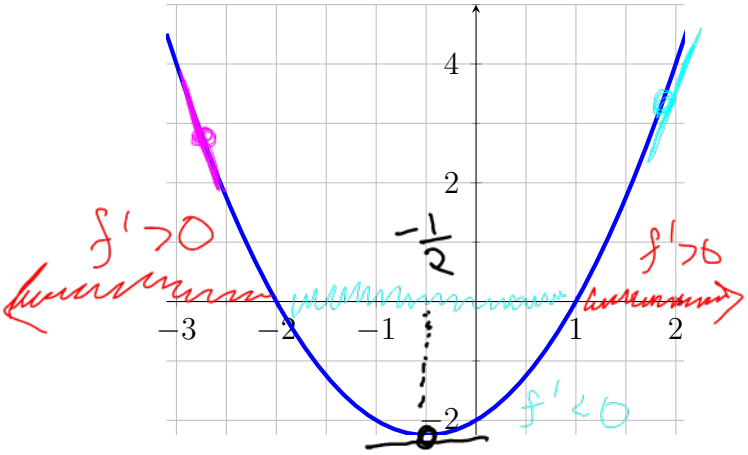
(c) State the locations of any relative max/mins as well as inflection points

Rel Min: None

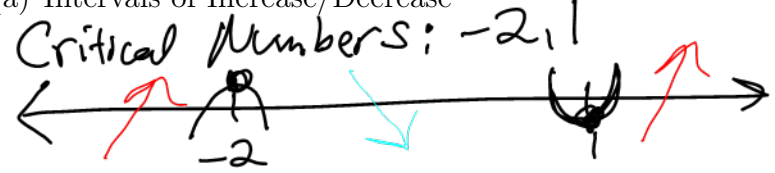
Rel Max: None

IPs: $x = -1, 0, 1$

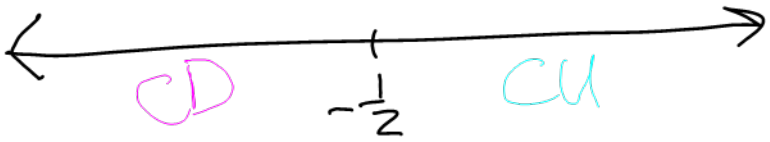
Problem 7. *Use the graph of f' to determine the following about f:*



(a) Intervals of Increase/Decrease



(b) When f is concave up VS concave down



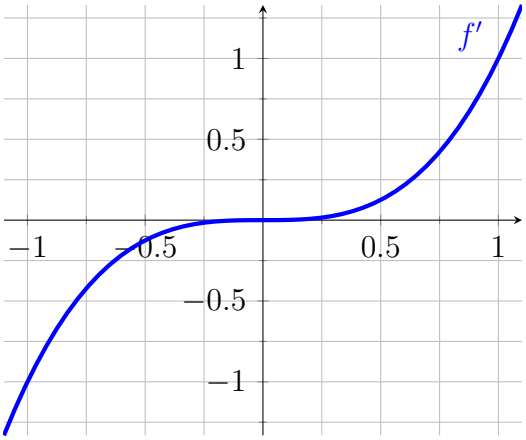
(c) State the locations of any relative max/mins as well as inflection points

Rel Min: $x=1$

Rel Max: $x=-2$

IPat $x=-\frac{1}{2}$

Problem 8. *Use the graph of f' to determine the following about f:*

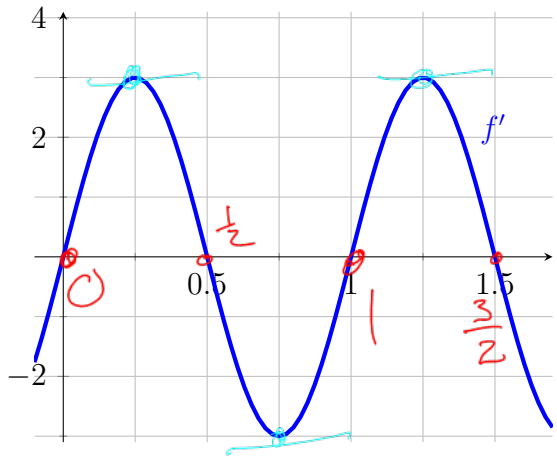


(a) Intervals of Increase/Decrease

(b) When f is concave up VS concave down

(c) State the locations of any relative max/mins as well as inflection points

Problem 9. *Use the graph of f' to determine the following about f:*



(a) Intervals of Increase/Decrease

(b) When f is concave up VS concave down

(c) State the locations of any relative max/mins as well as inflection points